Experiment No. 24 (Mixture No. 2)

Date: / /

Aim : Analyse two basic (cation) radicals qualitatively from given inorganic mixture.

Apparatus : Test tubes, test tube holder, test tube stand, filter paper etc.

A. Preliminary tests:

Test	Observation	Inference
Color	white	Ba^{2+}, Ba^{2+} may be present
Nature	crystalline	water sample salts present

B. Dry Tests for Basic radicals (Test tube must be dry for this test)

Test	Observation	Inference
1. Heating in a dry test tube:- Take a small quantity of the mixture a clean and in dry test tube and heat it strongly.	i) cracking noise ii) white residue which glows on heating	crystalline salts like $Pb(NO_3)_2$, NaCl, KBr may be Ba^{2+}, Sr^{2+}, Mg^{2+}, Ca^{2+} may be present
2. Charcoal Cavity Test: Mixture + Na_2CO_3 solid in 1:2 proportion placed in fresh charcoal cavity, moisten with a drop of water. Heat it with blow pipe in a reducing (yellow) flame	white residue	Al^{3+}, Ba^{2+}, Ca^{2+}, Mg^{2+} salts may be present
3. NaOH Test; Mixture + NaOH solution & heat. Hold moist turmeric paper near the mouth of the test tube	moist turmeric paper remains as it is	NH_4^+ absent
4. Flame Test: Prepare a paste of the given mixture with conc. HCl on a watch glass. Make a small loop at the end of the platinum wire & dip it in the mixture or use glass rod. Heat it on oxidising flame (Blue) observe the colour change of the flame.	Pale green	Ba^{2+} may be present

C. Preparation of original solution (O.S)

Take a small quantity of mixture in a beaker add 20 mL of distilled water, stir with glass rod to dissolve the mixture. If mixture does not dissolve completely then warm it to dissolve. Clear solution is obtained, which is used as a O.S for further tests.

1. Analysis of Group zero (NH_4^+)

Test	Observation	Inference
1. O.S. + NaOH solution + Heat, test with moist turmeric paper.	No evolution of NH_3 gas	Group zero is absent

2. O.S. + NaOH solution + Heat, Bring a glass rod dipped in conc. HCl. near the mouth of the test tube.	No dense white fumes of NH_4Cl	NH_4^+ is absent
C.T. for NH_4^+ : i. O.S. + Nessler's reagent in excess	—	—
ii. O.S + Picric Acid (2, 4, 6 trinitro phenol)	—	—

Detection of group (if group zero is absent, two groups must be detected following test)

Test	Observation	Inference
1. O.S. + dil. HCl	No white ppt	Group I absent
2. O.S./Filtrate + dil HCl (heat) + H_2S gas or water	No Black ppt	Group II absent
3. O.S./Filtrate (Remove H_2S) + NH_4Cl (equal) + NH_4OH (till alkaline to litmus)	no ppt	Group III absent
4. O.S./Filtrate + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + H_2S gas or water	No ppt.	Group-IV absent
5. O.S./Filtrate (Remove H_2S) + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) +$(NH_4)_2CO_3$	white ppt	Group V (Ca^{2+}, Ba^{2+}, Sr^{2+}) Present
6. O.S./Filtrate + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + Na_2HPO_4	White ppt.	Group VI (Mg^{2+}) Present.

2. Analysis of first detected group

Test	Observations	Inference
1) OS + NH_4Cl + NH_4OH + $(NH_4)_2CO_3$	white ppt	Ca^{2+}, Ba^{2+}, Sr^{2+} Present
2) above sol^n + K_2CrO_4	yellow ppt	Ba^{2+} Present

3. C.T. for first detected radical

Test	Observations	Inference
1. above acetate sol^n + dil H_2SO_4	white ppt	Ba^{2+} confirmed
2. above sol^n + $(NH_4)_2C_2O_4$	white ppt	Ba^{2+} confirmed

2. Analysis of second detected group Group VI

Test	Observations	Inference
1. group VI white ppt + dil. HCl	clear soln is obtained	Mg^{2+} present
2. —	—	—
3. —	—	—

3. C.T. for second detected radical Mg^{2+}

Test	Observations	Inference
1. above soln + Titan yellow soln	Rose red ppt	Mg^{2+} confirmed
2. above soln + Hypoiodite reagent	Reddish brown ppt	Mg^{2+} confirmed

Result:-

The given inorganic mixture no. 2 contains following two cations (Basic Radicals)

i) Ba^{2+} (Name of cation) Barium cation

ii) Mg^{2+} (Name of cation) Magnesium cation

Remark and sign of teacher: ....